

SIEMENS SIMOTICS GP 1LE1 STANDARD INDUCTION MOTOR
TECHNICAL SPECIFICATION DATASHEET - REF: 1LE1001-1DB2

1. GENERAL PRODUCT SPECIFICATIONS

Product Line: SIMOTICS GP Low-Voltage Motors
Model Identifier: 1LE1001-1DB23-4AA4
Motor Type: 3-Phase Squirrel-Cage Induction Motor
Frame Size: 160M (Cast Iron Frame)
Mounting Type: IM B3 (Foot-Mounted Horizontal)
Cooling System: IC411 (Self-Ventilated, Fan-Cooled)
Degree of Protection: IP55 Standard (Dust and Water Jet Protected)
Insulation Class: 155 (F), Temperature Rise Class: 130 (B)
Ambient Temperature Range: -20 C to +40 C (Standard Operation)

2. ELECTRICAL OPERATING RATINGS

Rated Output Power: 15.0 kW (20.0 HP)
Rated Operating Voltage: 400 V Delta / 690 V Star (+/- 10%)
Rated Frequency: 50 Hz (Optional 60 Hz Operation at 17.3 kW)
Rated Full Load Current: 28.5 A @ 400 V Delta
Rated Synchronous Speed: 1500 RPM
Rated Full Load Speed: 1475 RPM (2-Pole / 4-Pole Configuration)
Rated Full Load Torque: 97.1 Nm
Breakdown Torque Ratio: 3.1 x Rated Torque
Starting Current Ratio: 7.2 x Rated Current

3. ENERGY EFFICIENCY & STANDARDS

Efficiency Class: IE3 Premium Efficiency according to IEC 60034-30-1
Efficiency @ 100% Load: 92.1 %
Efficiency @ 75% Load: 92.4 %
Efficiency @ 50% Load: 91.8 %
Power Factor (cos phi) @ 100% Load: 0.83
Standards Compliance: IEC 60034, CE, UL, CSA, EAC Certified

4. MECHANICAL CHARACTERISTICS & BEARINGS

Drive End (DE) Bearing: 6309 2ZC3 (Deep Groove Ball Bearing)
Non-Drive End (NDE) Bearing: 6209 2ZC3
Lubrication: Pre-lubricated with high-temperature synthetic grease (40,000h service life)
Total Weight: 112 kg
Sound Pressure Level: 68 dB(A) @ 50 Hz full load
Terminal Box Position: Top-mounted (Rotatable 4 x 90 degrees)